

FIRST-ORDER LOGIC

Language	Ontological Commitment (What exists in the world)	Epistemological Commitment (What an agent believes about facts)
Propositional logic	facts	true/false/unknown
First-order logic	facts, objects, relations	true/false/unknown
Temporal logic	facts, objects, relations, times	true/false/unknown
Probability theory	facts	degree of belief $\in [0, 1]$
Fuzzy logic	facts with degree of truth $\in [0, 1]$	known interval value

Figure 8.1 Formal languages and their ontological and epistemological commitments.

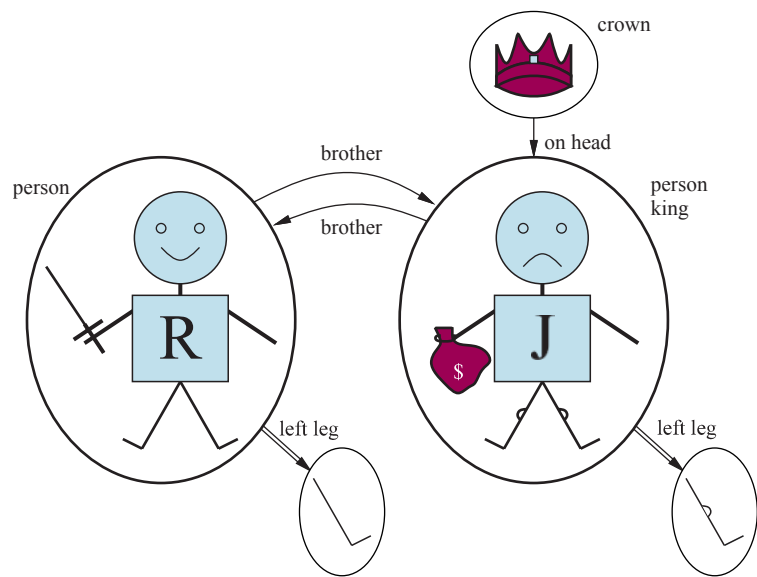


Figure 8.2 A model containing five objects, two binary relations (brother and on-head), three unary relations (person, king, and crown), and one unary function (left-leg).

$$\begin{aligned}
 \text{Sentence} &\rightarrow \text{AtomicSentence} \mid \text{ComplexSentence} \\
 \text{AtomicSentence} &\rightarrow \text{Predicate} \mid \text{Predicate}(\text{Term}, \dots) \mid \text{Term} = \text{Term} \\
 \text{ComplexSentence} &\rightarrow (\text{Sentence}) \\
 &\mid \neg \text{Sentence} \\
 &\mid \text{Sentence} \wedge \text{Sentence} \\
 &\mid \text{Sentence} \vee \text{Sentence} \\
 &\mid \text{Sentence} \Rightarrow \text{Sentence} \\
 &\mid \text{Sentence} \Leftrightarrow \text{Sentence} \\
 &\mid \text{Quantifier Variable}, \dots \text{Sentence} \\
 \\
 \text{Term} &\rightarrow \text{Function}(\text{Term}, \dots) \\
 &\mid \text{Constant} \\
 &\mid \text{Variable} \\
 \\
 \text{Quantifier} &\rightarrow \forall \mid \exists \\
 \text{Constant} &\rightarrow A \mid X_1 \mid \text{John} \mid \dots \\
 \text{Variable} &\rightarrow a \mid x \mid s \mid \dots \\
 \text{Predicate} &\rightarrow \text{True} \mid \text{False} \mid \text{After} \mid \text{Loves} \mid \text{Raining} \mid \dots \\
 \text{Function} &\rightarrow \text{Mother} \mid \text{LeftLeg} \mid \dots \\
 \\
 \text{OPERATOR PRECEDENCE} &: \neg, =, \wedge, \vee, \Rightarrow, \Leftrightarrow
 \end{aligned}$$

Figure 8.3 The syntax of first-order logic with equality, specified in Backus–Naur form (see page ?? if you are not familiar with this notation). Operator precedences are specified, from highest to lowest. The precedence of quantifiers is such that a quantifier holds over everything to the right of it.

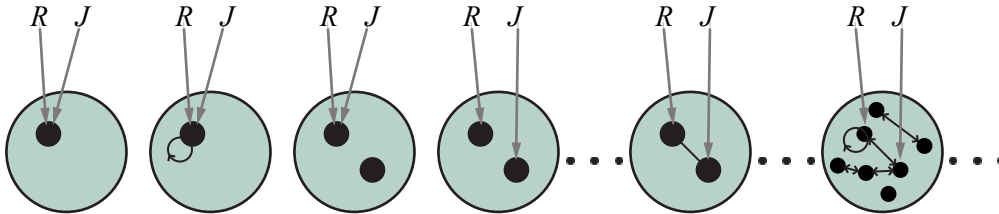


Figure 8.4 Some members of the set of all models for a language with two constant symbols, R and J , and one binary relation symbol. The interpretation of each constant symbol is shown by a gray arrow. Within each model, the related objects are connected by arrows.

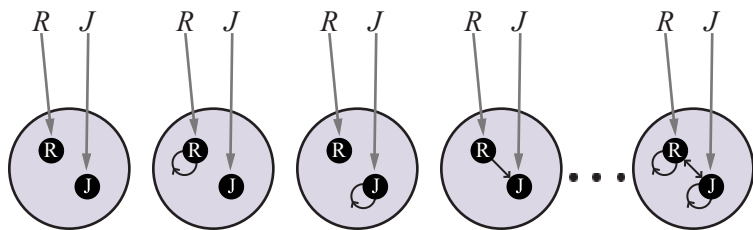


Figure 8.5 Some members of the set of all models for a language with two constant symbols, R and J , and one binary relation symbol, under database semantics. The interpretation of the constant symbols is fixed, and there is a distinct object for each constant symbol.

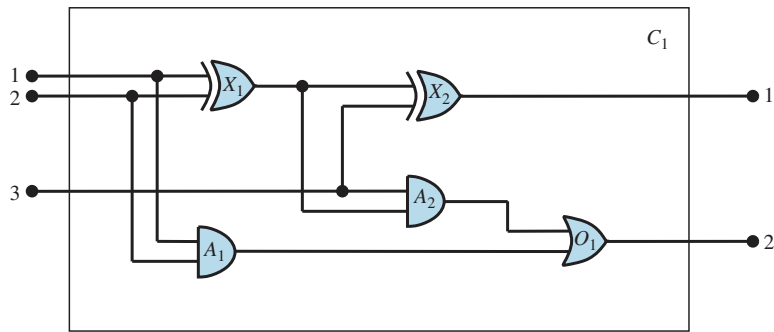


Figure 8.6 A digital circuit C_1 , purporting to be a one-bit full adder. The first two inputs are the two bits to be added, and the third input is a carry bit. The first output is the sum, and the second output is a carry bit for the next adder. The circuit contains two XOR gates, two AND gates, and one OR gate.